

AREA SURFACE AREA & VOLUME

Triangle:

- Perimeter of a triangle = Sum of all the three sides.
- Area of a triangle = $\frac{1}{2} \times \text{Base} \times \text{Height}$.
- Area of a triangle = $\sqrt{s(s-a)(s-b)(s-c)}$, where $s = \frac{a+b+c}{2}$.
- Perimeter of equilateral triangle = $3 \times \text{side}$.
- Area of equilateral triangle = $\frac{\sqrt{3}}{4} \times (\text{side})^2$.
- Height of equilateral triangle = $\frac{\sqrt{3}}{2} \times \text{side}$.
- Area of isosceles triangle = $\frac{1}{2} \times \text{Base} \times \sqrt{(\text{equal side})^2 - \frac{(\text{Base})^2}{4}}$.
- Height of isosceles triangle = $\sqrt{(\text{equal side})^2 - \frac{(\text{Base})^2}{4}}$.

Square:

- Perimeter of a square = $4 \times \text{side}$
- Area of square = $(\text{side})^2$.
- Diagonal of a square = $\sqrt{2} \times \text{side}$.

Rectangle:

- Perimeter of rectangle = $2 \times (\text{length} + \text{breadth})$.
- Area of rectangle = $\text{length} \times \text{breadth}$.
- Diagonal of rectangle = $\sqrt{(\text{length})^2 + (\text{breadth})^2}$.

Parallelogram:

- Area of a parallelogram = $\text{Base} \times \text{Height}$.

Rhombus:

- Area of rhombus = $\frac{1}{2} \times \text{Product of diagonals}$.
- Side of a rhombus = $\frac{1}{2} \sqrt{(d_1)^2 + (d_2)^2}$, where d_1, d_2 are lengths of diagonals.

Trapezium:

- Area of trapezium = $\frac{1}{2} \times \text{Sum of parallel sides} \times \text{Height}$

Circle:

- Circumference of a circle = $2\pi.r = \pi.d$.
- Area of circle = $\pi.r^2 = \frac{\pi.d^2}{4}$.
- Radius of circle = $\frac{\text{diameter}}{2}$.

Cube: Let side of the cube = a .

- Surface area of cube = $6a^2$.
- Area of four walls of cube = $4a^2$.
- Volume of the cube = a^3 .
- Main diagonal of a cube = $\sqrt{3} \times a$.

Cuboid: Let length = l , breadth = b and height = h

- Total surface area of cuboid = $2(lb + bh + hl)$
- Area of four walls of cuboid = $2h(l + b)$.
- Volume of the cuboid = $l \times b \times h$.
- Main diagonal of a cuboid = $\sqrt{l^2 + b^2 + h^2}$.

Cylinder: Let radius of base = r and height = h

- Base area of cylinder = $\pi.r^2$
- Curved surface area of cylinder = $2\pi.rh$
- Total surface area of cylinder = $2\pi.rh + 2\pi.r^2 = 2\pi.r(r + h)$.
- Volume of cylinder = $\pi.r^2h$

Cone: Let radius of base = r , height = h and slant height = l

- Slant height = $l = \sqrt{r^2 + h^2}$
- Base area of cone = $\pi.r^2$
- Curved surface area of cone = $\pi.rl$
- Total Surface area of cone = $\pi.r^2 + \pi.rl = \pi.r(r + l)$.
- Volume of the cone = $\frac{1}{3}\pi.r^2h$

Sphere:

- Curved surface area of sphere = $4\pi.r^2$
- Volume of the sphere = $\frac{4}{3}\pi.r^3$.

Hemisphere:

- Curved surface area of hemisphere = $2\pi.r^2$
- Base area of hemisphere = $\pi.r^2$
- Total surface area of hemisphere = $3\pi.r^2$
- Volume of the hemisphere = $\frac{2}{3}\pi.r^3$.

Frustum: Let h be height, l be the slant height and R and r are the radii of both the ends of a frustum.

- Slant height of frustum = $l = \sqrt{h^2 + (R - r)^2}$
- Smaller base area of frustum = $\pi.r^2$
- Bigger base area of frustum = $\pi.R^2$
- Curved surface area of frustum = $\pi.(R + r)l$.
- Total surface area of frustum = $\pi.r^2 + \pi.R^2 + \pi.(R + r)l = \pi.(r^2 + R^2 + Rl + rl)$.
- Volume of frustum = $\frac{1}{3}\pi.h(R^2 + r^2 + Rr)$.
